

SQUARE 40

Opening System — Full Technical Reference

Complete, reorganised transcription of the ALLUCO product catalogue

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1. Company & Product Range

ALLUCO provides solutions adapted to all high-end custom-made aluminium joinery projects. The company has an in-house design office working on 3D software, a workshop equipped with CNC machines, and experienced installation teams — ALLUCO designs, manufactures and installs, working directly with architectural projects.

Positioned on the aluminium market with innovative concepts adapted to contemporary architecture, ALLUCO works toward three goals: **(1)** improved energy performance of buildings (energy saving), **(2)** a healthy and comfortable home, and **(3)** a splendid design.

Catalogue pull-quote: "We not only develop innovative products and systems but also think about the aesthetic value of each building." Brand taglines: "success in progress" and "Your Partner For a Better Future", tied to ALLUCO's environmental commitment (see Section 14).

Full ALLUCO product range includes:

- Complete aluminium profile system series: sliding, door and opening, curtain wall
- Gates/portals (portails)
- Aluminium railings (garde-corps) for balconies, verandas, and aluminium-profile staircases — anodised inox-look or glazed finishes
- Extruded or injected roller-shutter slats for security
- Sun-protection systems: fixed or sliding sun-breakers (brise-soleil), pergolas, screens, camargue systems
- Ventilation grilles for healthy, ventilated spaces
- Thermal-break aluminium systems, environmentally respectful

SQUARE 40 is ALLUCO's 40 mm opening (swing / tilt-turn / French-window) aluminium system — offered with three frame variants: with reported joint cover, without joint cover, and with integrated joint cover.

2. Material — Aluminium Alloy & Extrusion

Raw material: 6000–6063 series aluminium alloy (Al, Mg, Si 0.5), state T5. Chemical composition conforms to standard ISO 209-1:1989 (F). Mechanical characteristics conform to standard ISO 6362-2.

Mechanical property	Minimum value
Hardness	68 HB
Breaking load	Rm = 200 MPa
Elastic limit	R0.2 = 160 MPa
Elongation	A = 10%

- Theoretical weights per linear metre shown in the catalogue are theoretical; actual weight is $\pm 10\%$.
- Standard bar length: 6.5 m (lacquered profiles).
- Plan an end-cut allowance of up to 10 cm per bar after surface treatment (lacquering) — mainly due to adhesion testing and hanging/handling marks.

3. Technical Characteristics & Performance

3.1 General characteristics

- Frame (dormant) depth: 40 mm
- Leaf (ouvrant) depth: 48 mm
- Reference overall installed section: from 89 mm
- Frame and leaf corners assembled with high-quality framing squares
- High-quality fittings, accessories and gaskets/joints — own manufacturing
- Straight and curved joint-cover (couvre-joint) design options
- Coplanar outdoor face on straight and curved sash lines (coplanaire à l'extérieur en ouvrant courbe et droit)
- Wide range of design options

3.2 Performance ratings

Performance	Rating	Standard
Air Permeability	Class 4	EN 12207
Water Sealing	6A	EN 12208
Wind Resistance	C4	EN 12210
Security (burglar resistance)	CR2	EN 1630

All four ratings are equal to or stronger than SQUARE 67's (Class 1 / 7A / C3 / CR2) except water sealing, reflecting SQUARE 40's role as an opening (swing/tilt-turn) system rather than a slider.

3.3 Opening types

Interior openings (ouvertures intérieures)

- Opening — Fenêtre à la française, 1 leaf (single French-style casement)
- Opening and tilt/turn — Oscillo-battant
- Tilt and turn — Basculant (bottom-hung / awning)
- Tilt and turn — Oscillo-battant (tilt-turn)

Exterior openings (ouvertures extérieures)

- Window — Fenêtre à l'anglaise, 1 leaf (outward-opening casement)
- Pliable — folding
- Opening — ouvrant à la française (outward French-style)

4. Dimensions & Glazing

Parameter	Value
Maximum glass thickness	27 mm
Medium theoretical glazing-bead thickness	1.35 mm
Maximum panel width (L)	2500 mm
Maximum panel height (H)	2400 mm
Maximum weight per leaf	120 kg *
Acoustic insulation	Rw = 32 dB (EN ISO 140-3 / EN ISO 717-1, areas ≤ 2.27 m ²)

* Always consult the exact weight and maximum dimensions for the specific configuration/hardware used.

5. Profile Range

Weights are theoretical (kg/ml, $\pm 10\%$ actual). "Perim." is the profile's external perimeter (mm), used for lacquering/surface-treatment calculations. $I(X)/I(Y)$ are moments of inertia (cm^4) where published. "Section (mm)" gives the overall cross-section envelope as dimensioned on the technical drawings.

5.1 Frames (Dormant)

Ref.	Designation (EN)	Section (mm)	Weight (kg/ml)	$I(X)/I(Y) \text{ cm}^4$	Perim. (mm)
FSQ 100	Single tubular frame	40 × 49.5 (72 tot.)	0.806	6.63 / 6.91	329
FSQ 102	Frame 'L' without joint cover	40 × 49.5, 27.5	0.622	3.64 / 5.84	249
FSQ 124	Single tubular 'L' frame, attached joint cover	40 × 49.5, 27.5	0.683	3.85 / 5.52	365
FSQ 148	Simple tubular frame, shape T	40 × 71.5, 26.5/27	0.764	6.61 / 6.33	414
FSQ 149	Double tubular 'L' frame, without joint cover	40 × 66.1, 44.1	0.809	9.53 / 7.84	299
FSQ 150	Double tubular 'Z' frame	40 × 66.1, 66.1	0.919	8.94 / 14.83	496
FSQ 151	Double tubular frame, shape T	40 × 88.6, 44.6	0.932	8.44 / 14.83	495
FSQ 165	Single tubular 'Z' frame, rounded joint cover 25mm	40 × 49.5, 52.5	0.803	— / —	326
FSQ 402	Single tubular 'Z' frame, rounded joint cover 35mm	40 × 49.5, 62.5	0.852	— / —	351
FSQ 408	Double tubular frame, added joint cover	40 × 66.1, 44	0.830	10.02 / 7.51	443
FSQ 535	Single tubular 'L' frame, weatherstripping + attached flap	40 × 49.5, 27.5	0.702	4.04 / 5.49	376
FSQ 536	Double tubular 'L' frame, weatherstripping + attached flap	40 × 66.1, 44.1	0.850	10.5 / 7.5	454

5.2 Sashes (Ouvrant)

Ref.	Designation (EN)	Section (mm)	Weight (kg/ml)	$I(X)/I(Y) \text{ cm}^4$	Perim. (mm)
FSQ 401	Single tubular opening (sash)	48 × 45, 36.3	0.867	10.10 / 7.85	459
FSQ 403	Double tubular opening	48 × 68.5	1.093	14.30 / 22.18	578
FSQ 404	Single glazed opening sash, F = 15 mm	48 × 61	0.877	9.63 / 7.434	463
FSQ 405	Double tubular 'T' opening	48 × 90.5, 46.5	1.094	13.43 / 22.18	578
FSQ 406	Double glazed opening window, F = 24 mm	48 × 61	0.843	9.51 / 6.54	445

5.3 Glazing Beads (Parclose)

Ref.	Designation (EN)	Section (mm)	Weight (kg/ml)	$I(X)/I(Y) \text{ cm}^4$	Perim. (mm)
FSQ 110	Straight glazing bead, F = 18 mm	18 × 22, 18.3	0.289	— / —	170
FSQ 111	Straight glazing bead, F = 24 mm	24 × 22, 12.3	0.284	— / —	164
FSQ 139	Straight glazing bead, F = 12 mm	12 × 22, 24.3	0.308	— / —	179

5.4 Intermediate Rails, Transoms & Beats

Ref.	Designation (EN)	Section (mm)	Weight (kg/ml)	$I(X)/I(Y) \text{ cm}^4$	Perim. (mm)
FSQ 104	Intermediate rail, 89 mm	88.6 × 88.6, 44.6	1.081	8.27 / 16.38	543
FSQ 121	Intermediate rail, 130 mm	40 × 130, 86	1.394	12.68 / 64.97	709

Ref.	Designation (EN)	Section (mm)	Weight (kg/ml)	I(X)/I(Y) cm ⁴	Perim. (mm)
FSQ 130	Beat inverter (inverseur de battement)	40 × 50.3	0.674	3.69 / 4.23	341
FSQ 112	Central beat (battement central)	40 × 61, 28.5	0.795	6.51 / 5.99	440
FSQ 156	Reinforced intermediate mullion/transom	70 × 40, 39/45	1.566	28.53 / 19.50	721

5.5 Corner, Cladding & Adapter Profiles

Ref.	Designation (EN)	Section (mm)	Weight (kg/ml)	I(X)/I(Y) cm ⁴	Perim. (mm)
FSQ 107	Rod (tringle)	19.5 × 2.5, 4.5	0.149	— / —	46
FSQ 108	Water discharge (rejet d'eau)	23.5 × 14.8, 22.5	0.137	— / —	75
FSQ 534	20 mm flange / flashing (bavette)	28.7 × 7	0.135	— / —	78
FSQ 131	Main cladding profile	15 × 100	0.813	12.21 / 29.13	491
FSQ 132	ANOD filler profile	50 × 10	0.348	2.19 / 3.01	215
FSQ 163	Corner profile, dia. 40 mm	dia. 40	0.645	4.32 / 4.32	238
FSQ 122	Adapter for gasket holder	22 × 38.57	0.301	— / —	169
FSQ 153	Frame assembly adapter, CSQ/FSQ	9.1 × 37	0.307	— / —	136
FSQ 164	Variable-angle assembly adapter	16.2 × 40, 11.5	0.288	2.39 / 2.21	164

5.6 Joint Covers (Couvre-Joints)

Ref.	Designation (EN)	Section (mm)	Weight (kg/ml)	I(X)/I(Y) cm ⁴	Perim. (mm)
CSQ 300	Round joint cover, 35 mm	35 (h)	0.183	— / —	109
CSQ 301	35 mm offset right joint cover	13.6 × 35	0.202	— / —	108
CSQ 302	35 mm right joint cover, to clip	8.6 × 35	0.178	— / —	108
CSQ 303	Round joint cover, 25 mm	9 × 25	0.116	— / —	89
CSQ 304	Round joint cover, 45 mm	13.03 × 45	0.178	— / —	128
CSQ 305	45 mm offset right joint cover	13.6 × 45	0.240	— / —	128
CSQ 306	45 mm right joint cover	8.6 × 45	0.216	— / —	118
CSQ 460	60 mm straight joint cover	20 × 60	0.380	— / —	210

6. Handles & Hinge Ranges

SQUARE 40 uses the same Supra7 minimalist handle family as SQUARE 67, plus a wider hardware range built around it: standalone Asia/Kora/NP handles, the Domina hinge and an invisible-hinge option ("Towards a new generation of handles — the innovative operating system that combines the aesthetics of a window handle and the simplicity of a cremone bolt").

Ref.	Designation
04153	Handle 'Supra 7 Asia' 7mm
04154	Handle 'Supra 7 Kora' 7mm
02415	Handle 'Asia' 7mm
00997	Handle 'Kora' 7mm
03960	Handle 'NP Ultra' 5mm (OB M180 simple tubular mechanism)
—	Supra NP / Supra Asia / Supra Kora — minimalist lever range

Hinges: Paumelle Bridge (door hinge, 00600N / 00120U for window), Paumelle Domina, Invisible hinge, Chic 3D 100 hinges (left/right, T1 and T2 leaf-length ranges), OB Classic 80 hinges, and False Hinge (04358, for symmetry on multi-point kits).

7. Hardware Kits

SQUARE 40 is supplied with complete, pre-defined hardware kits per opening type. Each kit below lists its components as printed in the catalogue, with the primary component and its "OU / Or" alternative where the catalogue offers a choice.

7.1 KIT PORTE 1 VANTAIL (1-leaf door kit)

Ref.	Designation	Qty
00600N	Bridge hinge for door	4
02431	ASIA rosette	2
SMSQR	1PT lock, base bolt, 3 square keys, stainless finish — OR	1
950302	3-point lock, flat strike with base bolt	1
GACHEALRE	Square strike, 2D inserts + fixing screws — OR	1
GACHEALRE 3PNT	3-point lock, flat strike with base bolt	1
02563	'Kora' lever handle — OR	1
02414	'Asia' lever handle	1

7.2 KIT PORTE 2 VANTAILS (2-leaf door kit)

Ref.	Designation	Qty
00600N	Bridge hinge for door	8
02144	Threshold socket with plug	1
03114K	Simple cremone strike plate, Zamak	1
02431	ASIA rosette	2
SMSQR	1PT lock (as above) — OR 950302	1
GACHEALRE	Strike (as above) — OR GACHEALRE 3PNT	1
02168K	Semi-fixed lock for door — OR	1
02191	PF bidirectional semi-fixed Futura lock	1
02563 / 02414	'Kora' or 'Asia' lever handle	1

7.3 KIT OF 1V VASISTAS / CLASSIC (1-leaf top-hung window)

Ref.	Designation	Qty
00120U	Window hinge	2
01704	Twin latch	1
02040K	150mm or 250mm limit-stop hinge for top-hung window	1
02041K	Kit component (top-hung actuator)	1
02574000K	Cremone kit (Classic variant)	1
00957 / 01076 / 00958	Cremone 'Kora' / 'Asia' / Keyed 'Kora'	1

7.4 KIT OF 1V / 2V CHIC VASISTAS (Chic top-hung)

Ref.	Designation	Qty
04362	Chic top-hung hinge kit component	1
01704 / 02574000K	Twin latch or Cremone kit	1
043552 / 043551	Chic hinges & supports 3D100, T1, left/right, leaf 470–699	1
043562 / 043561	Chic hinges & supports 3D100, T2, left/right, leaf 700–1500	1
02040K	150mm / 250mm limit-stop hinge	1

Ref.	Designation	Qty
00957 / 01076 / 00958	Cremone 'Kora' / 'Asia' / Keyed 'Kora'	1

7.5 KIT OF 1V / 2V CLASSIC — window cremone

Ref.	Designation	Qty
00120U	Window hinge	2–4
02574000K	Cremone kit	1
02191	PF semi-fixed bidirectional Futura lock (windows & French doors) — OR	1
02111K	OF GIAP semi-fixed lock (window)	1

7.6 KIT OF 1V / 2V CHIC 100 3D

Ref.	Designation	Qty
04352 / 043541	Chic hinges 3D100, left / right — 'Gauche ou droite'	1
0102001	Internal bidirectional simple tubular actuator	1
02167K	Adjustable Futura end cap	2
03114K	Simple cremone strike plate, Zamak	1
04153 / 04154 / 02415 / 00997	Handle: Supra7 Asia / Supra7 Kora / Asia / Kora, 7mm	1

7.7 KIT OF 1V / 2V NP 100 3D

Ref.	Designation	Qty
04352 / 043541	Chic hinges 3D100, left / right	1
03954 / 03960	French-style NP actuator / 'NP Ultra' 5mm handle (OB M180)	1
02167K / 03114K	Adjustable end cap / cremone strike plate	1–2

7.8 KIT OB 1V / 2V CLASSIC (tilt-turn, classic)

Ref.	Designation	Qty
04704	OB mechanism with cremone	1
04739	OB Classic 80 hinges	1–2
04339	Hinge type 1 OB, L 410–1300	1
04340	Hinge type 2 OB, L 550–1700	1
00959	Cremone 'Kora' OB, square	1
01084	Cremone 'Asia' OB	1
02191 / 02111K	Semi-fixed lock (as above), 2-leaf kit only	1

7.9 KIT OB 1V / 2V CHIC 100 3D (tilt-turn, Chic)

Ref.	Designation	Qty
0470501	OB M180 simple tubular mechanism	1
043552 / 043551	Chic hinges & supports 3D100 T1, left/right, leaf 470–699	1
043562 / 043561	Chic hinges & supports 3D100 T2, left/right, leaf 700–1500	1
04153 / 04154 / 02415 / 00997	Handle: Supra7 Asia / Supra7 Kora / Asia / Kora, 7mm	1
02191 / 02111K	Semi-fixed lock, 2-leaf kit only	1

7.10 KIT OB 1V / 2V NP 100 3D (tilt-turn, NP)

Ref.	Designation	Qty
0395501	OB M180 simple tubular mechanism	1

Ref.	Designation	Qty
03960	'NP Ultra' 5mm handle	1
043552 / 043551 / 043562 / 043561	Chic hinges & supports 3D100 (as above)	1
03114K	Cremone strike plate (2-leaf kit)	1

7.11 KIT SUPPLÉMENTAIRE (additional / multi-point parts)

Ref.	Designation	Qty
04358	False hinge	1
04770K	Additional lock	1
04301K	Additional hinge	1
04777K	Additional locking mechanism	1

7.12 KIT OF CHIC 1V À L'ITALIENNE (outward-opening, Italian style)

Ref.	Designation	Qty
01971	Polyamide filler block, groove 14/18 (frame)	4
01974	Polyamide filler block, groove 15/20 (active leaf)	4
02026	External opening kit	1
08534000	Italian compass GS HD CLPE — 10"	1
08536000	Italian compass GS HD CLPE — 14"	1
08538000	Italian compass GS HD CLPE — 16"	1
04105	Cremone kit 'Asia', for outward opening	1
01401	Cremone 'Asia', for outward opening	1

■ Kit component quantities above are reproduced as printed. Some catalogue pages mix reference numbers and quantities across adjacent columns (a scan/OCR limitation) — cross-check exact piece counts against the source page for the specific kit before ordering hardware.

8. Joints, Seals & Glazing Selection

8.1 Joint / seal references

Ref.	Designation
J5007 / J5006 / J5005 / J5004	Uniglass glazing gasket — 3 / 4 / 5 / 6 mm
J784	Central gasket
PJ64	Internal opening (battement) gasket
JB247	Opening gasket (rigid-base variant)
J242 / JV242	Exterior glazing gasket
J220	2 mm glazing gasket
JBR7X6 fin seal	6/7 mm brush gasket
ML02 – ML07	Minimal seal, 2 mm through 7 mm
03524590N	Minimal exterior glazing seal, 3 mm / connecting rod
J298 / J2034	Outer glazing gasket / opening gasket (bavette frame variant)

8.2 Glazing-bead / seal selection tables

Straight glazing bead, by glass thickness (outer gasket always ML03V)

Glass thickness (mm)	Glazing bead	Inner gasket
6–7	FSQ 139	ML02–ML03
8–9	FSQ 110	ML07–ML06
10–11	FSQ 110	ML05–ML04
12–13	FSQ 110	ML03–ML03
14–15	FSQ 111	ML07–ML06
16–17	FSQ 111	ML05–ML06
18–19	FSQ 111	ML03–ML04

Single-glazed opening sash FSQ 404 (F = 15 mm)

Glass thickness (mm)	Outer gasket	Inner gasket
6–7	ML05	ML04
8–9	ML04	ML03
10–11	ML03	ML02

Double-glazed opening window FSQ 406 (F = 24 mm)

Glass thickness (mm)	Outer/inner gasket
6–7	ML06 / ML06
8–9	ML05 / ML05
10–11	ML04 / ML04
12–13	ML03 / ML03

Volume-taking: glazing bead F=12/18/24mm — outer gasket J242

Bead	Glass thickness (mm)	Inner gasket
FSQ 139 (F=12mm)	4	J5005
FSQ 139 (F=12mm)	5	J5006
FSQ 139 (F=12mm)	6	J5007
FSQ 139 (F=12mm)	8	J220
FSQ 110 (F=18mm)	10	J5004

Bead	Glass thickness (mm)	Inner gasket
FSQ 110 (F=18mm)	12	J5005
FSQ 111 (F=24mm)	15	J5004
FSQ 111 (F=24mm)	16	J5005
FSQ 111 (F=24mm)	18	J5007
FSQ 111 (F=24mm)	20	J220

9. Composite / Corner Frame Assemblies (Chassis Composé)

For bay-window / corner configurations, SQUARE 40 uses a variable-angle corner post (FSQ 163 pivot profile + FSQ 164 variable-angle adapter), adjustable roughly between 140° and 30° depending on layout. Two documented cross-sections:

Section	Components
Horizontal section on corner post (standard frame FSQ102)	FSQ163, FSQ164, FSQ102, 36440FR, J784, PJ64, FSQ401, FSQ139, ML03V, ML03. Key dimensions: 16.2, 49.5, 5, 34, 11.6, 21.5, 45, 22 mm; angle range ≈140°–30°.
Horizontal section on corner post (bavette frame FSQ535/536)	FSQ163, FSQ164 (or FSQ401/FSQ111), FSQ534, FSQ535, FSQ536, 36440FR/36440SR, J784, J2102, PJ64, JB247, CSQ301. Same overall dimension set as the standard section, with the 20mm flange (FSQ534) added.
Vertical section through top beam	CSQ301, FSQ534, FSQ536/403/405, PJ64, J784, FSQ111, ML03/ML03V, key dimension 18 mm.

Note: "ML03"/"ML03V" are the 3mm minimal seal (interior/exterior); "JV220"/"J220" is the 2mm glazing seal.

10. Assembly Principles

Center flap assembly (Assemblage battement central)

Cap ACC40 112 fixes onto FSQ112 (central beat) and FSQ401 (sash), screwed from both faces.

Water nozzle / still-water frame assembly (Assemblage busette d'eau)

Plug ACC67 256 fits into frame profile FSQ102, machined with a 31mm slot set 6mm from the reference edge.

Frame assembly with bracket (Assemblage dormant avec équerre)

Corner bracket 36440FR joins mitred frame profile FSQ402 (or equivalent) at each corner.

Sash (opening) assembly with bracket (Assemblage d'ouvrant avec équerre)

Corner bracket 36440FR joins mitred sash profile FSQ401 at each corner, matching the frame assembly method.

Insertion of frame joints (Insertion des joints de dormant)

Seals J2102 and J784 are inserted into frame profile FSQ124 along its length.

Insertion of opening joints (Insertion des joints d'ouvrant)

Seals JV242 and PJ64 are inserted into sash profile FSQ401 at the corner.

Door plinth assembly (Assemblage de socle de porte)

Base profile FSQ121 and adapter FSQ122 are screwed to the door sash FSQ403 to form the bottom threshold detail — two documented variants for left/right hand configurations.

Handle/actuator mounting (machining ref. 01061000)

The OB actuator body is screwed to the sash profile (FSQ401/FSQ150/FSQ404/FSQ406) via a fixed drilling pattern: holes dia. 5.5mm and dia. 10.5mm spaced 21.5mm apart (43mm total span), positioned 81.85mm from each end on a 163.7mm total reference length.

11. Machining Reference Dimensions (Usage)

Key numeric callouts extracted from the CNC machining drawings (catalogue pages 80–81); the original dimensioned drawings remain the authoritative source for production tooling.

11.1 Machining for opening assembly square (sash corner bracket)

Key dimensions (mm): 11, 6, 22, 47, dia. 10.5.

11.2 Machining for frame assembly square (intermediate rail)

Key dimensions (mm): 9, 11, 47, dia. 10.5.

11.3 Machining for still-water frame assembly (water nozzle)

Slot 31mm, positioned 6mm from reference edge.

11.4 Machining for intermediate cross-member assembly

Key dimensions (mm): 22.25 (×2, symmetric about centreline), 4.

11.5 Handle/actuator drilling pattern (mechanism 01061000)

See Section 10 — holes dia. 5.5mm / dia. 10.5mm at 21.5mm pitch, 43mm span, 81.85mm end offsets, 163.7mm total; base plate 153.7mm inner span with 18mm end margins.

■ *These machining figures were transcribed from scanned dimensioned drawings and grouped by drawing rather than fully mapped point-by-point. Before cutting/CNC programming, cross-check against the original catalogue drawings (pages 76, 80–81) to avoid transcription error.*

12. Cutting Lists / Bill of Materials (Débitage)

The tables below give the profile cutting-length formulas (L = window/door width, H = overall height, h = leaf height where a threshold/mid-transom splits the opening, all in mm) and quantities for the four standard SQUARE 40 configurations, as published in the catalogue (pages 84–89).

12.1 1-Panel Opening Window (Fenêtre battante 1vtx)

Base frame FSQ124 (single tubular 'L', attached joint cover).

Profiles

Ref.	Designation	Cutting formula	Qty
FSQ124	Frame 'L', attached joint cover	L / H	2 / 2
FSQ401	Single tubular sash	L-44 / H-44	2 / 2
FSQ139	Straight glazing bead F=28mm	L-134 / H-178	2 / 2
FSQ107	Triangle (French window threshold)	(H-252)/2	1
FSQ108	Water discharge (inclined step)	L-111	1
CSQ301	35mm straight joint cover (mid transom)	L+62 / H+62	2 / 2

Accessories

Ref.	Designation	Qty
10840	Alignment bracket for sash	4
11840	Alignment bracket for frame	4
36440FR	Assembly bracket	8
001204	Hinge	2
00957	Cremone	1
ACC67 256	Busette (nozzle cap)	2
02574000K	Cremone bolt kit	1 kit
06501	Positioning cylinder	6

Joints

Ref.	Designation	Length formula
ML03V	Outer glazing gasket	2H+2L
ML03	3mm glazing gasket	2H+2L
J784	Central gasket	2H+2L
PJ64	Opening gasket	2H+2L

Glazing (glass cutting size)

Config.	Width formula	Height formula	Qty
Standard	L-149	H-149	1

12.2 2-Panel Opening Window (Fenêtre battante 2vtx)

Base frame FSQ124, with central beat FSQ112.

Profiles

Ref.	Designation	Cutting formula	Qty
FSQ124	Frame 'L', attached joint cover	L / H	2 / 2
FSQ401	Single tubular sash	(L-49)/2 / H-44	4 / 4
FSQ139	Straight glazing bead F=28mm	(L-229)/2 / H-178	4 / 4
FSQ107	Triangle (threshold)	(H-252)/2	2

Ref.	Designation	Cutting formula	Qty
FSQ108	Water discharge	$(L-183)/2$	2
FSQ112	Central beat	H-111	1
CSQ301	35mm joint cover (mid transom)	L+62 / H+62	2 / 2

Accessories

Ref.	Designation	Qty
10840	Alignment bracket for sash	8
11840	Alignment bracket for frame	4
36440FR	Assembly bracket	12
00120U	Hinge	4
00957	Cremone	1
ACC67 256	Busette	2
0257400K	Cremone bolt kit	1 kit
02111K	Semi-fixed lock kit	1 kit
ACC40112	Central-beat cap	1
06501	Positioning cylinder	8

Joints

Ref.	Designation	Length formula
ML03V	Outer glazing gasket	4H+4L
ML03	3mm glazing gasket	4H+4L
J784	Central gasket	3H+2L
PJ64	Opening gasket	4H+4L

Glazing (glass cutting size)

Config.	Width formula	Height formula	Qty
Standard	$(L-257)/2$	H-149	2

12.3 1-Panel Opening Door (Porte battante 1vtx)

Base frame FSQ408 (double tubular 'L', reported joint cover).

Profiles

Ref.	Designation	Cutting formula	Qty
FSQ408	Frame 'L', double tubular, joint cover	L / H	1 / 2
FSQ403	Double tubular sash	L-78 / H-46	1 / 2
FSQ111	Straight glazing bead F=24mm	L-215 / H-h-150 / H-266	4 / 2 / 2
FSQ121	Base profile	L-215	2
FSQ122	Base adapter	L-215	1
CSQ301	35mm joint cover (mid transom)	L+62 / H+31	1 / 2

Accessories

Ref.	Designation	Qty
10840	Alignment bracket for sash	2
11840	Alignment bracket for frame	2
36440FR	Assembly bracket	4
00600N	Bridge hinge	4
02563	'Kora' crutch handle	1
SMSQR	1PT vertical lock	1
GacheALRE	Square strike plate with insert	1
Vis 4.8×25 SIP	Assembly screw	8
06501	Positioning cylinder	6

Joints

Ref.	Designation	Length formula
ML03V	Outer glazing gasket	2H+4L
ML03	3mm glazing gasket	2H+4L
J784	Central gasket	2H+L
PJ64	Opening gasket	2H+L
JBR7X6 fin seal	Brush seal 7mm	2L

Glazing (glass cutting size)

Config.	Width formula	Height formula	Qty
Without mid-transom	L-227	H-236	1
With mid-transom FSQ121	L-227	H-h-160 / h-168	1 / 1
With mid-transom FSQ104	L-227	H-h-140 / h-147	1 / 1

12.4 2-Panel Opening Door (Porte battante 2vtx)

Base frame FSQ408, central beat FSQ112.

Profiles

Ref.	Designation	Cutting formula	Qty
FSQ408	Frame 'L', double tubular, joint cover	L / H	1 / 2
FSQ403	Double tubular sash	(L-83)/2 / H-46	2 / 4
FSQ111	Straight glazing bead F=24mm	(L-355)/2 / H-h-194 / H-202	8 / 4 / 4
FSQ121	Base profile	(L-357)/2	4
FSQ122	Base adapter	(L-357)/2	2

Ref.	Designation	Cutting formula	Qty
CSQ301	35mm joint cover (mid transom)	L+62 / H+31	1 / 2
FSQ112	Central beat	H-79	1

Accessories

Ref.	Designation	Qty
10840	Alignment bracket for sash	4
11840	Alignment bracket for frame	2
36440FR	Assembly bracket	4
00600N	Bridge hinge	6
02563	'Kora' crutch handle	1
SMSQR	1PT vertical lock kit	1 kit
GacheALRE	Square strike plate with insert	1
Vis 4.8×25 SIP	Assembly screw	16 kit
02168K	Semi-fixed lock (2nd leaf)	—
06501	Positioning cylinder	8

Joints

Ref.	Designation	Length formula
JV242	Outer glazing gasket	4H+4L
J5006	4mm glazing gasket	4H+4L
J784	Central gasket	3H+L
JB247	Opening gasket	4H+L
JBR7X6 fin seal	Brush seal 7mm	2L

Glazing (glass cutting size)

Config.	Width formula	Height formula	Qty
Without mid-transom	(L-385)/2	H-236	2
With mid-transom FSQ121	(L-385)/2	H-h-160 / h-168	2 / 2
With mid-transom FSQ104	(L-385)/2	H-h-140 / h-147	2 / 2

Both window configurations (12.1, 12.2) and both door configurations (12.3, 12.4) also have a documented **bavette** (weatherstripped flap) variant, substituting frame FSQ535 (single tubular) or FSQ536 (double tubular) for FSQ124/FSQ408, and adding profile FSQ534 (20mm flange, cut to L-4 / H-4 or H-8) plus seals J298 (outer glazing) and J2034 (opening) in place of the standard seal set.

13. Wind Load & Deflection

Design wind pressure: **600 Pascal** ($\approx 61 \text{ kg/m}^2$, $\approx 113 \text{ km/h}$). Maximum allowable deflection (arrow): **span / 300** (Flèche max 1/300). Two deflection curves are published, for two sash-reinforcement inertia values:

Reinforcement (Jx)	Curve shape	Plateau height (H, cm) at $B \geq 150\text{cm}$
$Jx = 10.1 \text{ cm}^4$	Starts $\approx H310$ at $B=30\text{cm}$, flattens from $B \approx 150\text{cm}$	≈ 200
$Jx = 14.3 \text{ cm}^4$	Starts $\approx H340$ at $B=30\text{cm}$, flattens from $B \approx 150\text{cm}$	≈ 210

Readings are approximate values taken from the published H(cm) vs B(cm) deflection charts (catalogue pages 92–93); consult the original charts for exact intermediate values. For window/door heights above 1800 mm, more hinges/locking points are recommended.

14. Certifications

- Profiles certified by the Italian laboratory Istituto Giordano ("certified by" mark, Beyond Certification) — ALLUCO SQUARE systems.
- "Nous nous engageons à préserver notre planète" / "We are committed to protecting our planet" — environmental commitment badge on current catalogue material.

15. Other ALLUCO Aluminium Systems

SQUARE 40 is part of ALLUCO's broader aluminium building systems range, which also includes:

- ALTO 15400 — swing/opening window system
- Oikos 4700 — swing/opening window system
- Supra 6000 — swing/opening window system
- Prima 6300 — swing/opening window system
- Klima 7400 & Klima PLUS — thermal-break window systems
- Entrance Doors MG90 — insulated entrance door system
- JEALOUSY — louvre / jalousie system
- Square 67 — sliding system (67 mm)
- Rolling Shutter systems
- Shadow Systems — sun-shading
- Curtain Wall systems
- UNION LINE — railings/balustrades
- ULTIMATE SKY — glass railings
- PURE LINE — glass railing/partition profiles
- Pergola AZORE

16. Contact

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